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SYSTEMATIC REVIEW

Does Acute Fatigue Negatively Affect Intrinsic Risk Factors of the Lower Extremity Injury Risk Profile? A Systematic and Critical Review

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Abstract

Background Acute fatigue is hypothesized to alter lower extremity injury risk profiles by affecting intrinsic risk factors (i.e. single leg postural control, hamstring strength). However, no systematic overview exists that merges the insights into prospective lower extremity injury risk profiling with the effect of acute fatigue on functional test performance.

Objective The objective of this review is to identify the influence of acute fatigue on prospectively determined modifiable intrinsic risk factors for lower extremity injuries.

Design Systematic review.

Methods PubMed (MEDLINE), Web of Science, PEDro, and Cochrane Library were searched until 29 May 2019. Studies were eligible when the study outcomes encompassed intrinsic modifiable risk factors for lower extremity injury, an acute fatigue intervention, and included healthy athletes or physically active people. Intrinsic modifiable risk factors were identified through recent systematic reviews and meta-analyses, and the referenced original research papers were used to determine outcome measures associated with increased injury risk.

Results Forty-three studies reported acute fatigue effects on modifiable risk factors, with eight studies matching all criteria for data-extraction. Acute fatigue can decrease single leg postural control, decrease ankle joint position sense, decrease isokinetic strength of hamstring and quadriceps muscles and can affect isokinetic hamstring:quadriceps ratios.

Conclusion Acute fatigue affects prospective intrinsic modifiable risk factors for lower extremity injury, indicating an altered injury risk profile for lateral ankle sprain, patellofemoral pain syndrome and hamstring injuries. Future research should allow for individual fatiguability as a relevant outcome, and merge insights from athlete-centred injury risk profiling and fatigue.

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Key Points

Acute fatigue can negatively affect intrinsic modifiable risk factors for lower extremity injuries and thus alter the injury risk profile.

Injury risk profiling should also be performed when the athlete is fatigued to approximate the sport context.

Individual fatigue responses should be considered when compiling an injury risk profile (i.e. % decline, fatigue index, absolute test decrement, etc.)

Future research needs to clearly delineate fatigue in order to assess possible underlying mechanisms and how these interact with injury risk.

1 Background

Preventing musculoskeletal injuries is crucial in sports and sports medicine research. Given the increasing body of evidence within the injury prevention domain over the past two decades, clinicians and researchers are able to look beyond singular research papers and can rely on literature reviews and meta-analyses to substantiate their decisions and actions. For example, recent reviews and meta-analyses documented various risk factors (i.e. single leg postural control, hamstring strength) for lower extremity injuries within different populations [1–16]. Combined, these review papers identified over 6500 research papers and included more than 125 studies in search of prospective risk factors for lower extremity injuries. Together, they provide insight into lower extremity injury risk profiles of athletes across different sports that transcend individual research papers. Although injury risk profiles cannot be used to predict injuries [17], they allow clinicians to perform evidence-based injury risk assessments and compile athlete-centred injury prevention plans. In order to ensure the quality and validity of such athlete-centred injury risk profiles, prospectively determined injury risk factors with a strong correlation between test outcome and injury risk should be included [17]. The recent UEFA injury study further illustrates the dire need for such multidisciplinary and athlete-centred approaches [18]. While the results of a declining incidence of some musculoskeletal injuries in recent years (i.e. ankle injuries) led to cautious optimism, the overall injury risk remained unchanged over the past 15 years and the number of hamstring muscle injuries per year continued to increase [18]. Such recent insights indicate that progress has been made over the past two decades within the injury prevention domain, but that many challenges remain to be overcome. One of these challenges is that contemporary injury risk profiles are currently lacking proximity to the sport-specific context [17–23].

An important constraint within this sport-specific context is the development of fatigue during athletic performance. Equally remarkable is the increased incidence of musculoskeletal injuries towards the end of a match [24]. This higher incidence rate of injuries during the last third of both practice and games is hypothesized to be associated with changes in neuromuscular control due to fatigue [25–27]. Fatigue can be considered as a complex phenomenon that manifests at both the central and peripheral level with continuous signal processing, feedback and feedforward loops [28–31]. Currently, no worldwide consensus exists on the definition of fatigue due to its complex and ubiquitous nature. Within the sport science domain, Enoka et al., Marcora et al., Gibson et al., Lambert et al., and Meeusen et al. defined and delineated different types of fatigue in the function of the associated mechanisms they studied [31–35]. However, within the sport context, multiple overlapping fatigue mechanisms are involved, advocating for a more general definition of fatigue at the behavioural level. Throughout this systematic review, we define fatigue according to Knicker et al. as an exercise induced decline of performance [36]. Remaining within the perspective of the sports and injury prevention domain, fatigue is also known to interact with an athlete's optimal performance [36, 37] and influence movement patterns in both healthy [38–40] and injured athletes [41–43]. Moreover, fatigue influences an athlete's functional test performance within the concepts of proprioception [44–46], balance and postural control [47–51], strength [52–55], and hop testing [56–59].

These changes in functional test performance are one of the main reasons why fatigue is hypothesized to be associated with an increased injury risk, while a sudden increase of acute workload, and perturbation of feedback loops, signal processing and feedforward loops are described as possible underlying mechanisms of this increased injury risk [60–62]. Furthermore, one has to consider the substantial intra- and interindividual differences in response to fatigue. This is why adding fatigue interventions to injury prevention screening tests and describing individual fatigue responses could be of great value, besides the regular injury prevention screening tests. Consequently fatigue would alter the injury risk of athletes in different ways and proportions. However, to date, no literature systematically merges the insights of prospective injury risk profiling for lower extremity injuries with the effects of acute fatigue on functional test performance. Therefore, the objective of this review is to identify the influence of acute fatigue on prospectively determined modifiable intrinsic risk factors for lower extremity injuries in a healthy, physically active population. The goal of this review is to support the understanding in how acute fatigue influences lower extremity injury risk profiles, and to determine if individual fatigue outcomes should be integrated within an athlete-centred injury risk profile and future injury prevention screening.

2 Methods

The review protocol was registered with the International Prospective Register of Systematic Reviews (PROSPERO) and developed in line with the Preferred Reporting Items for Systematic Review and Meta-analysis Protocols (PRISMA-P) [63]. The review protocol can be accessed via PROSPERO: https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=65274.

2.1 Information Sources

PubMed (MEDLINE), Web of Science, PEDro and Cochrane Library were searched for relevant articles from the dates of inception of these databases to 29th of May 2019.

2.2 Eligibility Criteria

Studies were eligible when the study outcomes encompassed intrinsic modifiable risk factors for lower extremity injury, an acute fatigue intervention, and included healthy athletes or physically active people. Fatigue was defined, in accordance with the work of Knicker et al., as an exercise induced impairment of performance [36]. Intrinsic modifiable injury risk factors were identified by recent systematic reviews and meta-analyses and the referenced original research papers were used to determine outcome measures associated with increased injury risk in specific populations (see Table 1) [1, 2, 4–16]. Together, the systematic reviews and meta-analyses evaluated over 125 studies that documented prospective risk factors for lower extremity injuries. The detailed overview of search results, population and injury risk outcome measures of the included systematic reviews and meta-analyses that documented the lower extremity injury risk profile can be found in the Electronic Supplementary Material (ESM) Table S1. Studies were excluded if the subject group comprised children or youth athletes (< 16 years old), elderly (65 + years old) or pregnant women. Only published articles in peer-reviewed journals were considered for full text inclusion. The scope of this review excluded kinematic measurements. Core stability and trunk proprioception outcomes were also excluded based on these being the focus of a recent review [64]. Exclusion criteria when screening full texts for data-eligibility were set as 'fatigue not standardized or clearly described' or when the outcomes measures were identical to the fatigue protocol (i.e. fatigue induced by repetitive hip abductions and test outcome being hip abduction strength). A summary of the eligibility criteria is presented in Table 2.

2.3 Search Strategy

The search strategy was developed by two authors (BT, JV) and encompassed three term sets that were created using the PICO method and were combined with 'AND'. The term sets included words for (P) 'healthy (athletic) population', (I) 'acute fatigue' and (O) 'injury risk factor' to ensure a sensitive search strategy. The words within each term were separated by 'OR'. The specialized and detailed search strategy for each database can be found in the ESM Table S2. Reference lists of relevant full text articles, reviews and meta-analyses were screened to identify possible additional articles.

2.4 Study Selection and Risk of Bias Assessment

The search was executed by one author (JV) across all selected databases. Next, all retrieved titles, abstracts, full texts and citations were integrated in the Rayyan web application (https://rayyan.qcri.org) [85] where publications were independently screened for inclusion by two authors (BT, JV) using a staged process of reviewing titles and abstracts, and eventually reviewing the remaining full texts. Results between the two researchers were compared, disagreements that were raised between the two researchers were resolved through discussion. If agreement could not be reached through discussion, a third author (RM) was consulted in order to make a decision. If no full text was available, corresponding authors were contacted or ordered via the library of the Vrije Universiteit Brussel. Full texts that matched all inclusion criteria were assessed using the Cochrane risk of bias assessment tool.

2.5 Data Extraction and Synthesis

During full-text screening, a detailed inventory was made of the acute fatigue protocols (ESM Table S3) and the effect of fatigue on modifiable intrinsic risk factors (see Table 3). To answer the research question, the extracted primary outcome measures comprised changes in test outcomes identified as significant intrinsic risk factors for lower extremity injuries as reported by the original prospective study designs (see Table 4).

3 Results

3.1 Study Collection

The systematic literature search yielded 1987 unique records in the selected databases, with 68 full-text articles that were eligible for data-extraction. Forty-three studies reported acute fatigue effects on modifiable risk factors, with eight studies matching all criteria for data-extraction (see Fig. 1). Sixty studies were excluded because they did not report the relevant outcomes that were identified as prospective injury risk factors by the original research papers.

3.2 Cochrane Risk of Bias Assessment

All eight included studies were screened with the Cochrane risk of bias assessment (see Figs. 2 and 3). Overall attrition and reporting bias were low. Selection, performance and detection bias were either unclear or not applicable to the study design.

Table 1 Modifiable intrinsic risk factors identified through systematic reviews and meta-analysis

Injury	Population	Modifiable intrinsic risk factor	Outcome measure	Original references
ITBS [1]	Healthy runners (male and female)	Decreased isometric hip muscle weakness	HHD—decreased torque (%BW h^{-1})	Fredericson et al. [65]
			HHD—decreased Newtons of force divided by body mass 2/3 HHD—decreased hip flexor and hip abductor strength HHD—increased hip adductor strength	Niemtuh et al. [66]
Non-contact ACL injury [2, 3]	Soccer players and athletes (male and female)	Decreased isokinetic H:Q ratios	Hcon: Qcon at 300° • s^{-1}	Myer et al. [67]
		Decreased isometric hip abduction and exorotational strength	HHD—(% BW)—decreased hip abductor and exorotational strength	Khayambashi et al. [68]
Hamstring injury [4–6]	Soccer players & athletes (male and female)	Decreased isokinetic H:Q ratios	Hcon: Qcon at 60° • s^{-1} and 240° • s^{-1}	Croisier et al. [69]
			Qcon: Hecc at 240° • s^{-1}/30° • s^{-1} Hcon: Qcon at 180° • s^{-1}	Croisier et al. [69]
			Hcon: Qcon at 60° • s^{-1}	Yeung et al. [70], Bennell et al. [71], Cameron et al. [72], Orchard et al. [73], Sugiura et al. [74]
		Bilateral differences in isokinetic testing > 15%	Qcon at 60° • s^{-1} and 240° • s^{-1} Hecc at 30° • s^{-1} and 120° • s^{-1} Hecc at 180° • s^{-1} Hecc at 60° • s^{-1} and 180° • s^{-1}	Croisier et al. [69], Fousekis et al. [75]
		Decreased hamstring peak torque Decreased hamstring mean normalised strength	Hcon at 60° • s^{-1}—decreased peak torque (Nm) Hecc at 60° • s^{-1}—decreased strength (Nm/Kg)	Orchard et al. [73] Sugiura et al. [74]
		Decreased hamstring flexibility	Active SLR test—degrees(°)	Henderson et al. [76]
			Passive SLR test—degrees (°)	Witvrouw et al. [77]
			Passive ROM—hip flexors and knee flexors (°) for general muscle strain in lower extremity	Bradley et al. [78]
LAS injury [7]	Athletes (male and female)	Decreased ankle joint position sense (proprioception)	Passive joint inversion position sense (15° • s^{-1}) Passive joint maximal inversion –5 position sense	Willems et al. [79]
		Decreased postural control (single leg balance)	Flamingo balance test—more attempts to keep balance	Willems et al. [80] Willems et al. [79]
			Neurocom Balance Master—decreased LOS directional control (%) Neurocom Balance Master—decreased LOS endpoint excursion (%) Neurocom Balance Master—decreased LOS max endpoint excursion (%) NeuroCom: composite score (°S/S): average degrees of sway per second—high sway	McGuine et al. [81]

Table 1 (continued)

Injury	Population	Modifiable intrinsic risk factor	Outcome measure	Original references
PFPS [8]	Adolescents and athletes (male and female)	Weaker quadriceps peak torque	Qcon at 60° • s^{-1} and 240° • s^{-1} —decreased absolute peak torque (Nm)	Duvigneaud et al. [82]
			Qcon at 60° • s^{-1}—decreased absolute and relative peak torque (Nm and Nm/kg, respectively) Qcon at 240° • s^{-1}—decreased absolute and relative peak torque (Nm and Nm/kg, respectively)	Van Tiggelen et al. [83]
Generic LE injury [9]	Athletes (male and female)	Altered isokinetic H:Q ratio	Hcon:Qcon peak torque at 90° • s^{-1}—lower = trauma; higher = overuse	Soderman et al. [84]

ITBS iliotibial band syndrome, *ACL* anterior cruciate ligament, *LAS* lateral ankle sprain, *PFPS* patellofemoral pain syndrome, *LE* lower extremity, *H:Q* ratio hamstring:quadriceps ratio, *HHD* handheld dynamometry, *BW* body weight, *Hcon* concentric hamstring contraction, *Qcon* concentric quadriceps contraction, *Hecc* eccentric hamstring contraction, *SLR* straight leg raise, *ROM* range of motion, *LOS* limits of stability

3.3 Impact of Fatigue on the Prospective Injury Risk Factors for Lower Extremity Injuries

Table 3 presents the acute fatigue effect on prospective injury risk outcome measures. Study characteristics of the eight included studies are shown in Table 4. The inventory of the acute fatigue protocols and the effect on modifiable risk factors ($n = 43$) can be found in ESM Table S3. Only 4 out of 8 studies reported the elapsed time between the completion of the acute fatigue protocol and the post-fatigue functional test assessment (see Table 4).

The effect of acute fatigue on isokinetic hamstring:quadriceps (H:Q) ratios that were prospectively identified as risk factors for hamstring injury was reported by 5 studies (see Table 3). Only one study reported a significant decrease in the specific concentric hamstring:quadriceps ratio (Hcon:Qcon) at 60° • s^{-1} [86] or the Hcon:Qcon at 180° • s^{-1} [87], while the other studies reported no effect of acute fatigue on these Hcon:Qcon ratios [88–90]. No study reported bilateral differences in isokinetic testing or the H:Q ratios related to non-contact anterior cruciate ligament (ACL) or generic lower extremity injuries. A significant decrease in isokinetic hamstring strength was reported by 2 studies [86, 88] with Table 3 also reporting a consistent decrease in isokinetic hamstring strength after acute fatigue. Results on the acute fatigue effect on quadriceps peak torque, peroneus longus reaction time, hamstring flexibility and ankle joint position sense (JPS) all show contradictory results (see Table 3). With regard to postural control and stability, 55 full-text articles were eligible for data-extraction, 29 reported single leg postural control, but no study selected the original outcome measures that were reported by Willems et al. or McGuine et al. [79–81]. However, Table 3 shows that acute fatigue predominantly decreases single leg postural control and stability. No data were available on isometric hip muscle strength.

3.4 Overview of Fatigue Protocols

Six studies used a running protocol to induce fatigue [25, 86–90] and 2 studies induced fatigue by using local fatigue protocols [91, 92]. Fatigue protocols of these eight studies are described in Table 4. Three studies used the Loughborough Intermittent Soccer Test (LIST) to induce fatigue [87, 88, 90]. The LIST is an intermittent exercise protocol simulating sport-specific demands. The three other running protocols used treadmill running to induce fatigue, being either a 90 min soccer specific intermittent protocol [86], a fixed duration protocol, based on energy expenditure [89] or an incremental running protocol until exhaustion [25].

The study of Mohammadi et al. [92] induced local ankle muscle fatigue using isokinetic dynamometry until subjects failed to reach 70% maximal voluntary contraction (MVC),

Table 2 Eligibility criteria

Inclusion	Exclusion
Healthy athletes or physically active people	Injured population (< 6 months)
Exercise induced fatigue Pre–post fatigue testing	Fatigue not standardized or clearly described Outcome measure identical to fatigue protocol
English Report at least one modifiable intrinsic risk factor (Table 1) with outcome measure identified as significant risk factors (Table 1)	Elderly (65 + years) Children or youth (< 16 years) Pregnant Disease (e.g fibromyalgia, multiple sclerosis, cancer) Kinematic measurements Core stability Trunk proprioception

while the study of Muyor et al. [91] induced local hip flexor muscle fatigue until subjects failed to reach an active range of motion (ROM) of 65° hip flexion.

4 Discussion

Acute fatigue negatively affects prospective injury risk factors that are part of the lateral ankle sprain (LAS), patellofemoral pain syndrome (PFPS) and hamstring injury risk profile, while insufficient data were available on iliotibial band syndrome (ITBS), non-contact ACL and generic lower extremity injury risk profiles. The principal findings of this review suggest that the body of evidence regarding the influence of acute fatigue on prospectively determined modifiable intrinsic risk factors for lower extremity injuries is scarce and of low quality. This limits our understanding on how acute fatigue interacts with injury risk. Although the need to focus on prospectively determined injury risk factors is clear [17], the effect of acute fatigue on these established risk factors often remains undocumented. Not one of the included studies in this systematic review reported the effect of acute fatigue on injury risk factors for non-contact ACL injuries [67, 68] as identified by Myer or Khayambashi, or on injury risk factors for ITBS as identified by Fredericson et al. [65] or Niemtuh et al. [66], or on injury risk factors for generic lower extremity injuries as identified by Soderman et al. [84].

4.1 The Influence of Acute Fatigue on the Lower Extremity Injury Risk Profile

When considering the hamstring injury risk profile, acute fatigue significantly decreases both concentric hamstring (Hcon) peak torque at 60° • s^{-1} and eccentric hamstring (Hecc) normalized strength at 60° • s^{-1} indicating an increased hamstring injury risk. Nevertheless, it seems that acute fatigue does not consistently alter the isokinetic H:Q ratios at Hcon:Qcon 60° • s^{-1} and Hcon:Qcon 180° • s^{-1}, while bilateral differences or other H:Q ratios remain undocumented with regard to fatigue. Also, hamstring flexibility as measured with the passive straight leg raise-test (PSLR) appeared to be influenced by acute fatigue. It is important to notice however, that the study of Muyor et al. found an increase in PSLR [91], while a decreased PSLR outcome predisposes to hamstring injury. One possible explanation could be that the results were not corrected for possible confounding factors (i.e. warm-up),which makes it difficult to attribute these results to the sole effect of fatigue. Based on these results, it appears that acute fatigue does affect the hamstring injury risk by decreasing hamstring peak torque and normalized strength with conflicting evidence about H:Q ratios. Nonetheless, more targeted research is necessary to further clarify this interaction.

LAS injury is one of the most often incurred lower extremity injuries [93–99], with documented high recurrent rates [100–104] and long lasting sensorimotor impairments [105–110]. Proprioception and postural control were identified as important concepts within the injury risk profile for LAS injury. Within the concept of proprioception, Mohammadi et al. [92] reported ankle JPS to be significantly impaired by fatigue, whereas Steib et al. found no difference in ankle JPS [25]. Although Steib et al. found a slight decrease in ankle JPS, one possible explanation for this contradiction in evidence could be that the study of Steib et al. [25] was underpowered, including 19 participants compared to 72 participants in the study of Mohammadi et al. [92]. Together with the acute fatigue effect on single leg postural control, this inhibition of ankle JPS due to fatigue can negatively affect LAS injury risk and could be an important constraint within the LAS injury risk profile.

The influence of acute fatigue on the prospective PFPS injury risk profile was documented by three included studies [86, 87, 89]. All three studies reported a post-fatigue decrease in quadriceps peak torque at 60° • s^{-1}, with two studies reporting this difference as a significant decrease [86]. No data on fatigue and quadriceps peak torque at 240° • s^{-1} were reported. Based on these findings, acute fatigue

Table 3 Acute fatigue effect on modifiable intrinsic risk factors and outcome measures

Modifiable risk factor	Acute fatigue effect on modifiable risk factor (ESM Table S2)	Prospective injury risk outcome measure	Included studies reporting outcome measure (n =)	Acute fatigue effect on prospective injury risk outcome measure
Decreased isometric hip muscle weakness	No data available	HHD—decreased torque (%BW h^{-1}) HHD—decreased Newtons of force divided by body mass 2/3 HHD—decreased hip flexor and hip abductor strength HHD—increased hip adductor strength	0	???
Decreased isokinetic H:Q ratios	↑↑↑ ~ ○○○○○○○○○ ↓↓↓ ●●●●●	Hcon: Qcon at 60° • s^{-1}	4	↓↓↓ [86], ~ [87, 89, 90]
		Hcon: Qcon at 90° • s^{-1}	0	???
		Qcon: Hecc at 240° • s^{-1}/30° • s^{-1}	0	???
		Hcon: Qcon at 180° • s^{-1}	3	↓↓↓ [87], ~ [89, 90]
		Hcon: Qcon at 300° • s^{-1}	0	???
Decreased isometric hip abduction and exorotational strength	No data available	HHD—(% BW)—decreased hip abductor and exorotational strength	0	???
Bilateral differences in isokinetic testing > 15%	No data available	Qcon at 60° • s^{-1} and 240° • s^{-1} Hecc at 30° • s^{-1} and 120° • s^{-1} Hecc at 180° • s^{-1} Hecc at 60° • s^{-1} and 180° • s^{-1}	0	???
Decreased hamstring isokinetic strength	↑↑↑ ~ ↓↓↓ ●●●●●●	Hcon at 60° • s^{-1}—decreased peak torque (Nm)	1	↓↓↓ [86]
		Hecc at 60° • s^{-1}—decreased strength (Nm/Kg)	1	↓↓↓ [88]
Decreased hamstring flexibility	↑↑↑ ● ~ ○○○ ↓↓↓	Active SLR test—degrees(°)	0	???
		Passive SLR test—degrees (°)	1	↑↑↑ [91]
		Passive ROM—hip flexors and knee flexors (°)	0	???
Decreased ankle JPS (proprioception)	↑↑↑ ~ ○○ ↓↓↓ ●●	Passive joint inversion position sense (15° • s^{-1})	2	↓↓↓ [92], ~ [25]
		Passive joint maximal inversion –5 position sense	1	↓↓↓ [25]
Decreased postural control (single leg balance)	↑↑↑ ~ ○○○ ↓↓↓ ●●●●●●●●●●●●●●●●●●●●●●●●●●●●●●●●	Flamingo balance test—more attempts to keep balance	0	???
		Neurocom Balance Master—decreased LOS directional control (%)		
		Neurocom Balance Master—decreased LOS endpoint excursion (%)		
		Neurocom Balance Master—decreased LOS max endpoint excursion (%)		
		NeuroCom: composite score (°S/S): average degrees of sway per second—high sway		
Weaker quadriceps peak torque	↑↑↑ ~ ○○ ↓↓↓ ●●●●	Qcon at 60° • s^{-1}—decreased absolute and bodyweight normalized peak torque (Nm and Nm/kg)	3	↓↓↓ [86, 87], ~ [89]
		Qcon at 240° • s^{-1}—decreased absolute and bodyweight normalized peak torque (Nm and Nm/kg)	0	???
Increased functional LSI in hop testing	↑↑↑ ~ ↓↓↓ ●	/	/	/
Delayed peroneus longus reaction time	↑↑↑ ●● ~ ○ ↓↓↓ ●	/	/	/

H:Q ratio hamstring:quadriceps ratio, *LSI* Limb Symmetry Index, *JPS* joint position sense, *SLR* straight leg raise, *HHD* Hand held dynamometry, *%BW* percentage of body weight, *LOS* limits of stability, *ROM* range of motion, *Hcon* concentric hamstring contraction, *Qcon* concentric quadriceps contraction, *Hecc* eccentric hamstring contraction, ??? no data available; ~ no difference, ↓↓↓ significant decrease, ↑↑↑ significant increase, ○ number of studies reporting no difference, ● number of studies reporting a significant difference, / = no significant risk factor in original research

does influence the PFPS injury risk profile by decreasing quadriceps peak torque at 60° • s^{-1}, but it remains unclear what the impact of acute fatigue is on the other modifiable intrinsic risk factors for PFPS.

The systematic reviews for non-contact ACL injury risk [2, 3] did not report single leg postural control as a modifiable risk factor. Although some debate exists about the interaction between single leg postural control and non-contact ACL injury risk, the results of the relevant reviews are confirmed by a more recent prospective study that was unable to identify static and dynamic postural control outcomes as prospective risk factors [111]. Poor postural control is suggested to play a role in non-contact ACL injury mechanism [112–115]. Often this association is founded on video-analysis of the inciting event that demonstrates postural control to be diminished when landing or performing cutting manoeuvres. When prospectively evaluating dynamic and static postural control as a risk factor, this risk factor was not retained by the included review [2, 3] and no association was found in elite handball players [111]. This highlights the importance of prospectively determined injury risk factor outcomes with a proximate relation to injury occurrence. Apparently, observations made during the inciting event (i.e. poor postural control during cutting manoeuvre) do not translate to the prospective injury risk profile (i.e. single leg postural control) of the susceptible athlete. Table 3 and ESM Table S2 support the hypothesis that single leg postural control is negatively affected by acute fatigue, without specific data available on non-contact ACL-injury risk or LAS-injury. Again, more targeted research is needed to further clarify this interaction.

4.2 Towards Understanding the Role of Acute Fatigue in the Injury Risk Profile

Even though fatigue affects neuromuscular lower limb functioning [42], core muscle functioning [64] and single leg landing kinetics and kinematics related to non-contact ACL injuries [116, 117], fatigue does not uniformly affect these outcomes. Therefore, it is not possible to generalize these findings or to know which pathways play an important role when these alterations occur. Our review reinforces these findings by documenting the negative effect of acute fatigue on modifiable intrinsic risk factors (see Table 3). Of the eight included studies, 6 studies used a running protocol to induce a more general state of fatigue [25, 86–90], while 2 studies opted for a more local state of fatigue [91, 92]. With regard to the intensity and objectivation of the eight included fatigue protocols, one study let participants perform a fatigue protocol until subjective exhaustion [25], while the other studies used fatigue protocols with either a fixed time [86–88, 90], fixed caloric expenditure [89], or a predetermined physiological fatigue level that had to be reached in order to end the fatiguing task [91, 92]. Nevertheless, fatigue is a complex phenomenon, integrating both physiological and psychological components [118]. This is why these one-dimensional approaches to fatigue do not only limit our ability to understand underlying fatigue mechanisms, but also how fatigue interacts with and affects the injury risk profile. It is acknowledged that we need to move away from these one-dimensional definitions and objectivation strategies of fatigue, and accept complex regulation of human exercise performance and fatigue [33, 119].

Therefore, a multidimensional approach can advance our insights into how fatigue interacts with and affects the injury risk profile. One example of such a multidimensional approach to fatigue, integrating complex network models and correlations between interacting components, can be found in the work of Pereira et al. [120]. Within this multidimensional perspective of fatigue, future studies should integrate multiple physiological and psychological fatigue outcomes in order to objectify fatigue interventions and their corresponding mechanisms. These fatigue outcomes should also be selected in accordance with the study rationales and fatigue hypotheses [31–36]. Likewise it is important to further validate the psychological component of fatigue [31, 118] and to integrate insights into how mental fatigue could affect the injury risk profile [37]. In summary, fatigue should always be clearly defined and objectified in a multidimensional way in order to allow for a better understanding of the role of fatigue within the injury risk profile. Also the hypothesized peripheral (i.e. motor unit recruitment, energy depletion) or central mechanisms (i.e. feedback and feedforward loops and corresponding signal processing) by which fatigue and injury risk interact should be further explored.

4.3 Putting the Athlete First—Athlete-Centred Pattern Recognition and Individual Fatigue Responses

Complex systems theory, pattern recognition and the athlete-centred approach are becoming increasingly important within clinical practice as well as in sport injury prevention research. Bahr critically reviewed the use of individual injury risk factors and screening test outcomes in order to predict injury and concluded that this reductionist approach is inadequate to serve this purpose [17]. Within a complex system, consequences at the behaviour level cannot be attributed to the alteration in an individual component of the system, but rather explained by altered interactions between several underlying components of the system, leading to different outcomes [20–22, 121]. This is why an athlete can be considered as a complex system, with multiple underlying components interacting and an injury as an unwanted outcome of the system [20–22, 121]. Within the injury prevention domain, complex systems theory has led to new injury

Table 4 Study characteristics of the eight included studies

Study	Subject characteristics	Study design	Fatigue protocol	Fatigue objectivation	Outcome measurements	Results
De Abreu Camarda et al. [86]	$N=21$ male soccer players *Unbalanced group:* $N=10$, Age: 23.2 ± 3.8 years, Weight: 74.3 ± 7.0 kg, Height: 179.0 ± 7.0 cm Hcon:Qcon < 0.60 (range: 0.45–0.59) *Balanced group:* $N=11$, Age: 23.1 ± 3.1 years, Weight: 72.9 ± 7.6 kg, Height: 178.0 ± 6.0 cm Hcon:Qcon > 0.60 (range: 0.61–0.70)	PreT–fatigue–PostT	Soccer specific intermittent protocol on treadmill. High-intensity exercise bouts were separated by low-intensity recovery bouts. Total time = 90 min. + 15 min. rest	No	Isokinetic H:Q ratio 1) Maximal concentric contractions for Q and H on Biodex isokinetic dynamometer (Biodex System 3, Biodex Medical Systems, Shirley, NY) at 60° s^{-1} and 180° s^{-1} 2) Maximal eccentric contractions for Hecc at 180° s^{-1}	Qcon 60° s^{-1} peak torque decrease = sign. ($p<0.05$) Qcon 180° s^{-1} peak torque decrease = NS Hcon 60° s^{-1} peak torque decrease = sign ($p<0.05$) Hcon 180° s^{-1} peak torque decrease = NS Hecc 180° s^{-1} peak torque decrease = only sign. in balanced group ($p<0.05$) Hcon:Qcon 60° s^{-1} peak torque decrease = only sign. in balanced group ($p<0.05$)
Cohen et al. [88]	$N=9$ male soccer players Age: 25.3 ± 0.8 years, Weight: 77.0 ± 3.7 kg, Height: 178.8 ± 2.9 cm	PreT–fatigue–PostT	90-minute LIST	No	Isokinetic H:Q ratio Five maximal concentric and eccentric contraction for Q and H on an isokinetic dynamometer (HUMAC NORM, Computer Sports Medicine International, Stoughton, MA, USA) at 120° s^{-1}	Hecc 120° s^{-1} peak torque decrease = sign. ($p<0.05$)
Coratella et al. [90]	$N=22$ male soccer players Age: 20.1 ± 2.4 years	PreT–fatigue–PostT	90-minute LIST	No	Isokinetic H:Q ratio Three maximal concentric and eccentric contractions for Q and H on an isokinetic dynamometer (Cybex Norm, Ronconcoma, USA) at 60° s^{-1}, 180° s^{-1}, and 300° s^{-1}	Hcon:Qcon 60° s^{-1} = NS Hcon:Qcon 180° s^{-1} = NS
Delextrat et al. [87]	$N=8$ male soccer players Age: 21.3 ± 2.3 years; Weight: 78 ± 9 kg Height: 178 ± 8 cm	PreT–fatigue–PostT	90-minute LIST	Yes: HR Bla RPE	Isokinetic H:Q ratio Three maximal concentric contractions for Q and H, and eccentric contractions for H on an isokinetic dynamometer (Cybex Norm, Phoenix Healthcare, Nottingham, UK) at 60° s^{-1}. Five maximal concentric contractions for Q and H, and eccentric contractions for H on an isokinetic dynamometer (Cybex Norm, Phoenix Healthcare, Nottingham, UK) at 180° s^{-1}.	Hcon:Qcon 60° s^{-1} = NS Hcon:Qcon 180° s^{-1} = sign. ($p<0.05$) Qcon peak torque 60° s^{-1} = sign. ($p<0.05$)

Table 4 (continued)

Study	Subject characteristics	Study design	Fatigue protocol	Fatigue objectivation	Outcome measurements	Results
Mohammadi et al. [92]	$N=72$ male soccer players *Intervention group* $N=36$ Age: 24.7 ± 1.3 years; Weight: 78.9 ± 7.9 kg Height: 183.7 ± 8.2 cm; *Matched control group* $N=36$ Age: 23.9 ± 0.9 years; Weight: 77.8 ± 6.5 kg Height: 181.2 ± 6.9 cm	PreT–fatigue// control–PostT	Maintenance of workload equal to 70% of their MVC on a Biodex System 3 isokinetic dynamometer (Biodex Medical Systems) Control: 45-minutes soccer match	Yes: Exercise termination when subjects could no longer maintain 50% of MVC for ± 3 s	Ankle JPS Ankle JPS was repeated 3 times on an isokinetic dynamometer (Biodex System 3, Biodex Medical Systems)	Passive joint inversion position sense decrease (15° s^{-1}) for both groups = sign. ($p<0.05$) Passive joint maximal inversion –5 position sense decrease for both groups = sign. ($p<0.05$) Between groups = NS
Muyor et al. [91]	Total $N=75$ recreationally active adults Male $N=38$, Age: 24.1 ± 5.5 years, Weight: 67.8 ± 13.5 kg Height: 169.2 ± 6.1 cm Female $N=37$ Age: 22.4 ± 2.5 years Weight: 65.2 ± 12.5 kg, Height: 169.7 ± 7.7 cm	PreT–fatigue PostT	Fatigue of hip flexor muscles. Actively lifting the legs alternatively, to an angular value of 65° of hip flexion, as many times as possible.	Yes: Inability to perform more than one active hip flexion, lying supine	Hamstring flexibility passive straight leg raise (PSLR), active straight leg raise (ASLR), passive knee extension (PKE), active knee extension (AKE), sit-and-reach (SR) toe-touch (TT)	ASLR = sign. decrease ($p<0.05$) PSLR and PKE = sign. increase ($p<0.05$) AKE, SR and TT = NS
Oliveira Ade et al. [89]	$N=16$ physically active males Age: 22 ± 2.6 Weight: 79.6 ± 10.3 kg, Height 173.8 ± 27.9 cm VO_2: 45.1 ± 6.42 ml/kg/min.	PreT–fatigue// control–PostT	Continuous treadmill running at 95% OBLA, with total duration being estimated to a caloric expenditure around 500 kcal. Control: standardized warm-up period (5 min. cycling at 70 W)	Yes: Bla HR	Isokinetic H:Q ratio Five maximal concentric and eccentric contractions for Q and H on an isokinetic dynamometer (Biodex System 3, Biodex Medical Systems, Shirley, New York) at 60° s^{-1} and 180° s^{-1}.	Hcon:Qcon 60° s^{-1} = NS Hcon:Qcon 180° s^{-1} = NS Qcon peak torque 60° s^{-1} = NS
Steib et al. [25]	$N=19$ non-injured participants Male/Female: 13/6, Age: 23.3 ± 3.8 years, Weight: 71.2 ± 10.0 kg, Height: 175.9 ± 7.5 cm	PreT–fatigue–PostT	Treadmill running until subjective exhaustion: 1° of inclination Start at 8 km h^{-1}, Speed increased stepwise by 2 km h^{-1} every 3 min.	Yes: RPE ≥ 17 ("very hard")	Ankle JPS Tested twice in three different angles of ankle inversion (10°, 15°, 20°) on an isokinetic dynamometer (Biodex 2, Biodex Medical Systems Inc, Shirley, New York)	Passive JPS = NS

N number, *PreT* pre-test, *PostT* post-test, *H:Q* ratio hamstring:quadriceps ratio, *Q* quadriceps, *H* hamstrings, *Qcon* concentric quadriceps contraction, *Hecc* eccentric hamstring contraction, *Hcon* concentric hamstring contraction, *LIST* Loughborough Intermittent Soccer Test, *MVC* maximal voluntary contraction, *JPS* joint position sense, *HR* heart rate, *Bla* blood lactate, *RPE* rating of perceived exertion, *OBLA* onset of blood lactate accumulation, *sign.* significant, *NS* not significant

prevention models and changed the focus from risk factor identification towards pattern recognition [20, 21]. In this context, fatigue can be either considered as a sole factor within the system (i.e. athlete) or as an underlying interacting component in the system's behaviour. Consequently, the centre of attention in future research should not comprise the functional performance test outcome or isolated risk factors, but should interpret the results from an athlete-centred approach. Within this perspective, exploring fatiguability of the individual athlete level could be a more relevant outcome than significant group differences. For instance, the study of Oliveira Ade et al. reported significant post-fatigue group differences, as well as a wide variability in relative change within post-fatigue injury risk test outcomes of fatigue [89]. These results highlighted very wide standard deviations, indicating a very individual fatigue response. A recent study with special focus on fatiguability also documented that match-play fatigue alters landing biomechanics. It was shown that the fatigue protocol did not cause overall alterations in landing patterns. However, when fatiguability was taken into account, high fatiguability was related to less optimal landing patterns that are associated with an increased injury risk [122]. Methodologically, fatiguability was considered on a subjective scale, measuring rating of perceived exertion (RPE). These individual fatigue responses can be monitored over time and serve pattern recognition if integrated in a continuous or repeated injury risk monitoring approach. Possible outcomes for fatiguability include subjective ratings of fatigue, together with outcomes related to the exercise induced impairment of performance (i.e. % decline, fatigue index, absolute test decrement, etc.). Once again, it is crucial to define and objectify fatigue and to measure the hypothesized fatigue mechanisms within this athlete-centred approach. Eventually, this will allow us to better understand the complex nature of injury occurrence and fatigue as a constraint within the athlete-centred injury risk profile. Therefore, individual fatigue outcomes should be integrated within an athlete-centred injury risk profile and future injury prevention screening.

From the perspective of on-field clinicians, our review reinforces the statement and popular belief that functional performance tests are affected by acute fatigue and can only provide a temporal snapshot of the current state of the injury risk profile [123, 124]. Accepting the temporal nature of test outcomes, implies the need for a more continuous data gathering strategy, pattern recognition and injury risk monitoring over time. Coaches, athletes, physiotherapists and doctors should therefore always consider test outcomes within the total athlete perspective and the relevant context in which these functional performance tests belong. Fatigue is indisputably part of this relevant context [36]. Although future research should focus on identifying and validating more practical solutions, we can advise clinicians and coaches working with athletes that functional testing and injury risk profiling should integrate fatigue in the performance and injury risk assessment protocol. In research, we need to standardize both fatigue and the assessment of injury risk outcome measures, but on-field clinicians can translate this to a more specific type of fatigue coupled to the functional performance tests that they consider relevant in the performance environment. These individual pre- and post-fatigue performance data, monitored over time during the season, can provide valuable individual insights and possibly expose patterns that cannot be identified by functional tests alone. Furthermore, these emerging insights into complexity and injury risk profiling allow us to better understand why screening for injury did not work [17] and why we need to focus on high quality prospective research identifying injury risk factors. If we merge the athlete-centred approach (i.e. allow for fatiguability), the temporal nature of functional performance tests, and combine them with the previously stated insights into fatigue (i.e. a complex phenomenon, define and objectify the hypothesized peripheral and mechanisms) [31–36], we can mutually progress our understanding in injury risk profiling and injury occurrence. The goal of merging these concepts is to keep athletes healthy and ready to perform. Furthermore, this paradigm shift towards an athlete–centred approach could also be applied to secondary and tertiary prevention, within rehabilitation progress and return to sport decisions [125] or to long term neural adaptations following injuries [126, 127].

4.4 Limitations

Given the absence of relevant published articles concerning risk factors that predispose for non-contact ACL, ITBS or generic lower extremities injuries, it is unclear to what extent fatigue influences these injury risk profiles. The injury risk profile for lower extremity injuries, as documented by recent systematic reviews and meta-analyses [1–16] appears to be founded upon a limited amount of original research papers [65–84]. With regards to limb symmetry index in hop testing (recurrent hamstring injury) and peroneal muscle reaction time (LAS injury), inconsistencies were found between the original findings of the research papers and the reporting of the result by these systematic reviews and meta-analyses. When consulting the original research papers the injury risk factors were not significant [128, 129], while the systematic reviews and meta-analyses did report a significant difference. The study of Patrek et al. was excluded because the fatigue protocol encompassed local hip abductor muscle fatigue and the outcome was isometric hip muscle strength [130]. Furthermore, the active straight leg raise data were not extracted from the study of Muyor et al. because the aim of the fatigue protocol was to induce fatigue of hip flexors [91]. It was also not possible to match the included

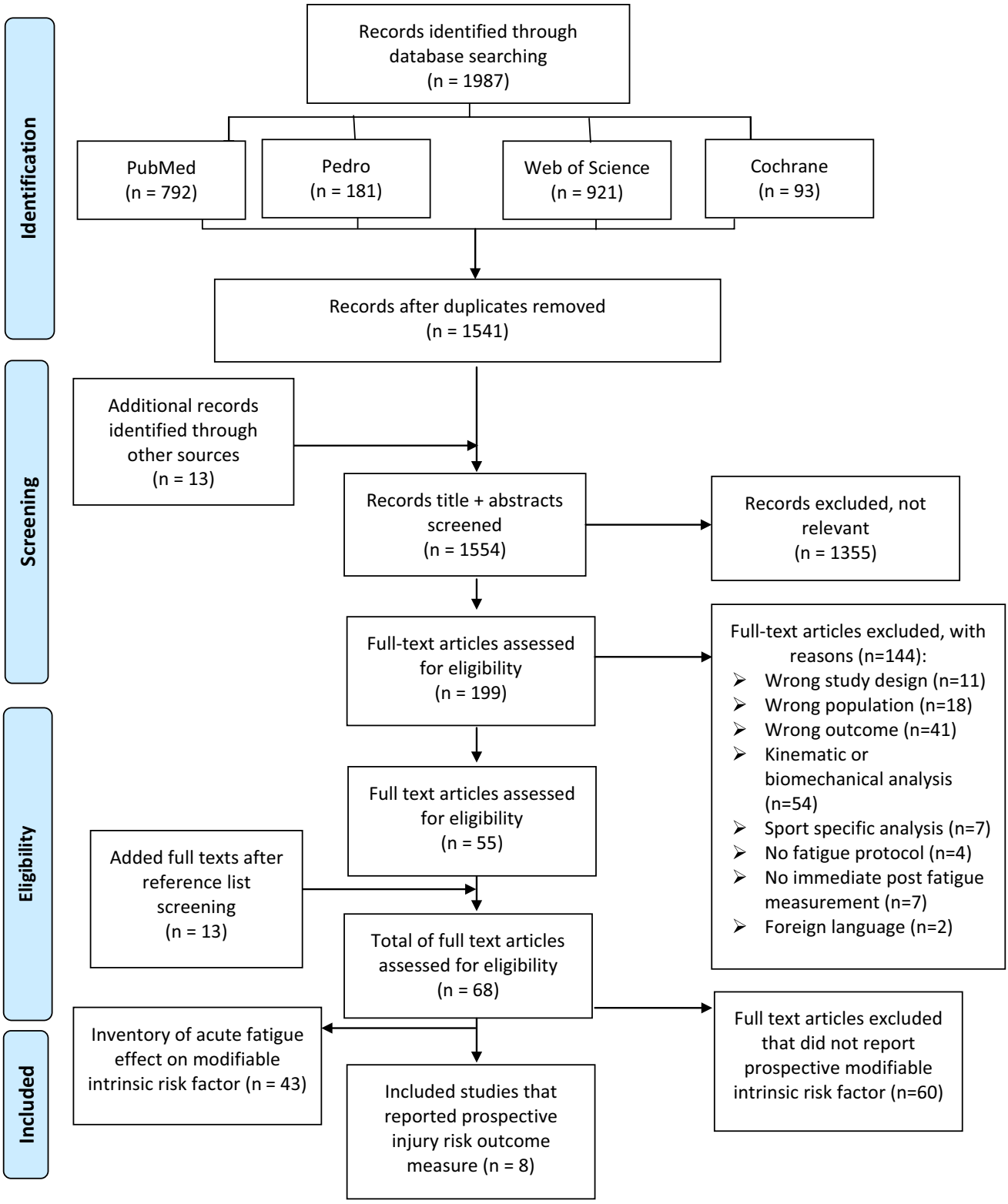
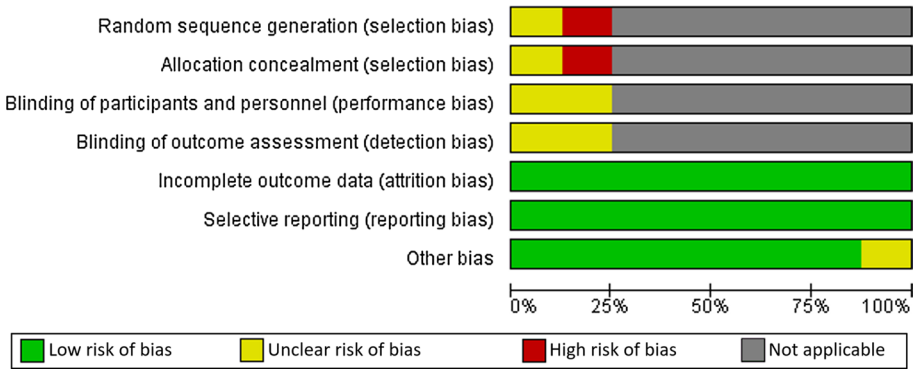


Fig. 1 Study identification and selection flowchart

Fig. 2 Risk of bias graph



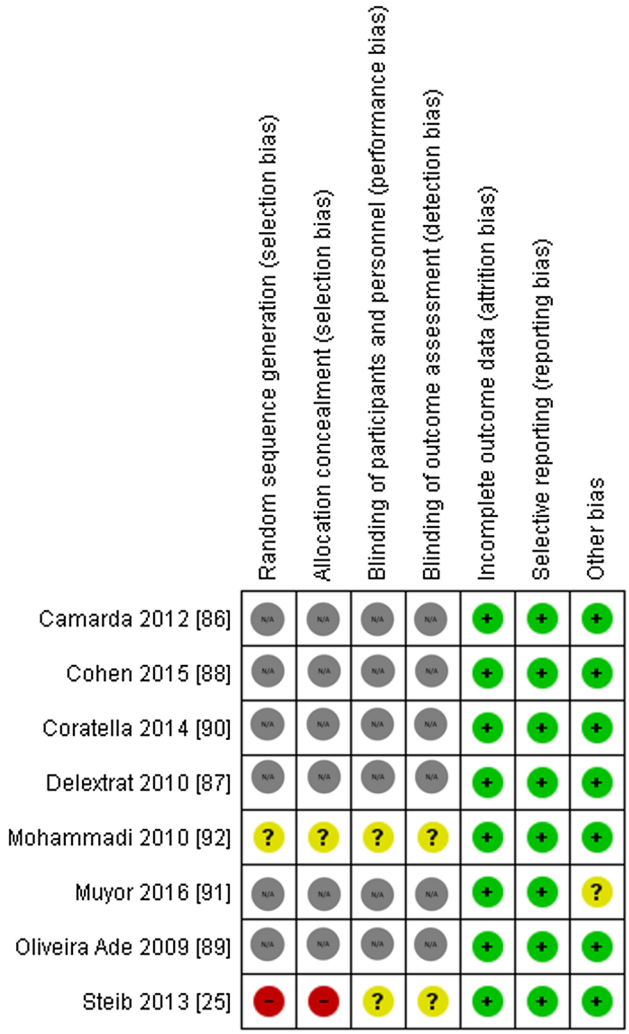


Fig. 3 Risk of bias summary

population with the population included in the reviews or original research papers that document lower extremity risk factors. The population of the included papers predominantly consisted of male participants, with only the study of Steib et al. including just 7 women [25]. The data should thus be interpreted in this perspective and cannot be generalized across other populations. Readers can consult ESM Table S1 for a more specific overview of included populations. Only 4 out of 8 included studies reported the elapsed time between the end of the acute fatigue intervention and the following functional test assessment. This highlights the importance of future high quality prospective research documenting the lower extremity injury risk profile. Another limitation of this systematic review is that a meta-analysis could not be performed due to the limited data available, high risk of bias of included papers and the risk that merging interventions would lead to unrecognized heterogeneity (i.e. incongruence between fatigue protocols, incongruence between populations, limited number of studies reporting the same outcome). Given the current evidence, it is not indicated to generate an overall estimate of effect size of acute fatigue in relation to prospective injury risk factors for lower extremity injuries. For example, the fatigue protocol of the 2 studies examining ankle JPS are an incremental running protocol [25] and a 70% MVC local isokinetic dynamometer protocol [92]. On the other hand, if one would consider to perform a meta-analysis on the effect of acute fatigue on the included H:Q ratios, the research question would be different and more aligned with for example hamstring injury risk. In such case, the meta-analysis would benefit from a more specific search strategy and possibly including similar fatigue protocols from the ESM Table S3 inventory of fatigue protocols. With the current review, we aim at initiating high quality research which will allow researchers to perform a qualitative meta-analysis in the future.

Finally, we acknowledge that other prospective risk factors have been identified for lower extremity injuries that were not included in the systematic reviews and meta-analyses. Researchers should continuously develop this injury risk profiling domain with high quality prospective study designs. This will allow for updated systematic reviews and meta-analyses and allow us to make a more inclusive and specific injury risk profile in the future. Finally, within a complexity view on injury prevention and fatigue, we accept the possibility that there are other, currently unknown

intrinsic risk factors that relate to lower extremity injuries that were not discussed in this review.

4.5 Future Perspectives for Research and Clinical Practice

Future research should consider the use of continuous or periodical injury risk profiling, with the inclusion of fatiguability. In a performance environment, functional testing and injury risk profiling should integrate fatigue in the performance and injury risk assessment protocol. A first step could be to use sport-specific fatigue protocols in this setting and document individual fatigue response in sport-specific and functional tests. These individual pre- and post-fatigue performance data, monitored over time, could possibly expose patterns that cannot be identified by functional tests alone. Possible outcome measures for fatiguability can include both subjective (i.e. RPE) and objective outcomes (i.e. % decline, fatigue index, absolute test decrement, etc.). In research, the use of longitudinal research designs that allow for big data monitoring in designing injury risk profiles are warranted. To promote further insight into fatigue, it is crucial to define and delineate the fatigue protocol and the hypothesized fatigue mechanisms that interact with test outcomes on an individual level. Cooperation in research is warranted when merging these insights into injury risk profiling, fatigue and fatiguability. The combination of cooperation and advancing insights could allow us to better understand the complex nature of injury risk profiling and injury occurrence.

5 Conclusion

Evidence on the influence of acute fatigue on prospectively determined modifiable intrinsic risk factors for lower extremity injuries is scarce. Acute fatigue affects the LAS, PFPS and hamstring injury risk profile, with insufficient data available on ITBS, non-contact ACL and generic lower extremity injury risk profiles. Our results suggest that acute fatigue can negatively affect prospectively determined injury risk factors, causing decreased isokinetic H:Q ratios, isokinetic hamstring and quadriceps peak torques, ankle JPS and single leg postural control, as well as a decreased functional limb symmetry index in hop testing and conflicting results for peroneus longus reaction time. With the current state of evidence, we are unable to substantiate these statements, although they are often hypothesized. We hope that our systematic review can contribute to promote more high quality research within the fatigue and injury risk research domain. Future research should aim to explore to what extent prospectively identified injury risk factors are affected by fatigue, allow for individual fatiguability as a relevant outcome measure, and merge insights from athlete-centred injury risk profiling and fatigue to better understand the complexity of injuries and the adaptive nature of injury risk profiles.

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Compliance with Ethical Standards

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Data availability statement The authors declare that all data supporting the findings of this study are available within the article [and its supplementary information files].

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